
Unit – 3 : Session – 4 : SLO – 1 | FCFS and SJF Programs — Missing C Code (FCFS)

Answers

Waiting time: $wt[i] = wt[i-1] + bt[i-1]$;

Turnaround time: $tat[i] = wt[i] + bt[i]$;

Completed program

```
#include <stdio.h>
int main() {
    int n, bt[20], wt[20], tat[20], i;
    float avg_wt = 0, avg_tat = 0;

    printf("Enter total number of processes: ");
    scanf("%d", &n);
    printf("Enter process burst times:\n");
    for(i = 0; i < n; i++) {
        printf("P[%d]: ", i+1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i-1] + bt[i-1];           // TODO: waiting time
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];             // TODO: turnaround time
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        printf("P[%d]\t%d\t\t%d\t\t%d\n", i+1, bt[i], wt[i], tat[i]);
    }
    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
    return 0;
}
```